Day 030 | Unit 02 | Teacher plan

Positive Rational Numbers: Fractions and Mixed Numbers

Unit 2 Test

Standards: 6.NS.B.3, 6.NS.B.2, 6.NS.A.1

Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

enVision Topic 1: Use Positive Rational Numbers | Spectrum companion Chapter 2

Assessment routine

5 minutes: explain response types without solving any item. 35 minutes: independent assessment. 10 minutes: collect work and note misconceptions for the next lesson. Keep the answer key and worked-example help unavailable during assessment.

Review allocation

Questions 1–4 are previous-unit review (20%). Current-unit questions increase in demand after the review. Unit 1 uses Grade 5 readiness skills because no earlier course unit exists.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 9, 12, 15

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Directions

The first 4 questions (20%) review Unit 1. The remaining questions assess this unit. Work in order and show your reasoning. Each whole item receives equal credit; a multi-select item requires every correct selection and no incorrect selection.

20 questions | 100 points: 5 per question

01 / Previous unit review | Constructed response

Use the standard algorithm: 20.8 + 8.7. Show your regrouping or division steps.

02 / Previous unit review | Constructed response

Use the standard algorithm: 35.3 - 1.5. Show your regrouping or division steps.

03 / Previous unit review | Constructed response

Use the standard algorithm: 93.3 × 8.4. Show your regrouping or division steps.

04 / Previous unit review | Constructed response

Use long division: 350 ÷ 7. Show the algorithm and check.

05 / Constructed response

Calculate 1/4 × 1/4. Simplify.

06 / Constructed response

Calculate 2 1/4 × 2 1/2. Simplify.

07 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 8 parts. Circle groups of 1 parts. Use the model to find 1 ÷ 1/8.

08 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 3 parts. Circle groups of 2 parts. Use the model to find 3/8 ÷ 1/4.

09 / Constructed response

Draw a bar partitioned into 8 equal parts. Shade 4 parts. Circle groups of 3 parts. Use the model to find 1/2 ÷ 3/8.

10 / Constructed response

Calculate 8/5 ÷ 1/4. Simplify.

11 / Constructed response

Calculate 3/4 ÷ 1/5. Simplify.

12 / Multiple choice

Calculate 3 1/6 ÷ 1 1/4. Simplify. Select one answer, then justify it with a model or calculation.
A. 15/38
B. 23/15
C. 38/15
D. 53/15

13 / Multiple choice

Calculate 4 1/3 ÷ 2 3/4. Simplify. Select one answer, then justify it with a model or calculation.
A. 85/33
B. 52/33
C. 19/33
D. 33/52

14 / Multiple choice

Calculate 1/2 ÷ 2/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 4/3
B. 7/4
C. -1/4
D. 3/4

15 / Model and constructed response

The equation 3/5 × n = 7/3 represents equally sized portions. Find n, write the related division equation, and check.

16 / Model and constructed response

A jug contains 1/3 L. Each full bottle needs 2 L. How many full bottles can be filled, and how much liquid remains?

17 / Model and constructed response

A jug contains 1 L. Each full bottle needs 1/2 L. How many full bottles can be filled, and how much liquid remains?

18 / Multiple choice

Calculate 2/3 ÷ 2/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 2
B. 1
C. 0

19 / Multiple choice

Calculate 1/2 × 1/2. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 1/4
B. -3/4
C. 4
D. 5/4

20 / Multiple choice

Calculate 2 3/4 × 1 1/3. Simplify and explain why your answer is reasonable. Select one answer, then justify it with a model or calculation.
A. 3/11
B. 11/3
C. 14/3
D. 8/3

Day 030 | Unit 02 | Answer key

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Learning target: I can choose a model, solve accurately, explain my reasoning, and check the result.

1. 29.5

Align decimal points; 20.8 + 8.7 = 29.5.

6.NS.B.3 | Constructed response

2. 33.8

Align decimal points; 35.3 - 1.5 = 33.8.

6.NS.B.3 | Constructed response

3. 783.72

Compute the whole-number product and restore decimal places: 93.3 × 8.4 = 783.72.

6.NS.B.3 | Constructed response

4. 50

35 ÷ 7: write 5, subtract 35, remainder 0; bring down the next digit: 0 ÷ 7: write 0, subtract 0, remainder 0. Check: 7 × 50 = 350.

6.NS.B.2 | Constructed response

5. 1/16

1/4 × 1/4 = 1/16.

6.NS.A.1 | Constructed response

6. 45/8

9/4 × 5/2 = 45/8.

6.NS.A.1 | Constructed response

7. 8

Each small part is 1/8; 8 shaded parts divided into groups of 1 gives 8/1 = 8 groups.

6.NS.A.1 | Constructed response

8. 3/2

Each small part is 1/8; 3 shaded parts divided into groups of 2 gives 3/2 = 3/2 groups.

6.NS.A.1 | Constructed response

9. 4/3

Each small part is 1/8; 4 shaded parts divided into groups of 3 gives 4/3 = 4/3 groups.

6.NS.A.1 | Constructed response

10. 32/5

8/5 ÷ 1/4 = 8/5 × 4 = 32/5.

6.NS.A.1 | Constructed response

11. 15/4

3/4 ÷ 1/5 = 3/4 × 5 = 15/4.

6.NS.A.1 | Constructed response

12. C - 38/15

19/6 ÷ 5/4 = 19/6 × 4/5 = 38/15.

6.NS.A.1 | Multiple choice

13. B - 52/33

13/3 ÷ 11/4 = 13/3 × 4/11 = 52/33.

6.NS.A.1 | Multiple choice

14. D - 3/4

1/2 ÷ 2/3 = 1/2 × 3/2 = 3/4.

6.NS.A.1 | Multiple choice

15. n = 35/9; 7/3 ÷ 3/5 = 35/9

7/3 ÷ 3/5 = 7/3 × 5/3 = 35/9. Check: 3/5 × 35/9 = 7/3.

6.NS.A.1 | Model and constructed response

16. 0 bottles; 1/3 L left

1/3 ÷ 2 = 1/3 × 1/2 = 1/6. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

17. 2 bottles; 0 L left

1 ÷ 1/2 = 1 × 2 = 2. Only whole bottles count; subtract the liquid used.

6.NS.A.1 | Model and constructed response

18. B - 1

2/3 ÷ 2/3 = 2/3 × 3/2 = 1.

6.NS.A.1 | Multiple choice

19. A - 1/4

1/2 × 1/2 = 1/4.

6.NS.A.1 | Multiple choice

20. B - 11/3

11/4 × 4/3 = 11/3.

6.NS.A.1 | Multiple choice